

Day 2 – SQL Server Schema Design & Normalization

1. Project Entities

The system contains the following entities:

- Customers
 - Categories
 - Products
 - Orders
 - OrderItems
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2. Database Tables

Customers

Column	Data Type	Key
Id	INT	PK
Name	NVARCHAR(100)	
Email	NVARCHAR(255)	
Phone	NVARCHAR(20)	
Address	NVARCHAR(255)	
CreatedAt	DATETIME2	

Categories

Column	Data Type	Key
Id	INT	PK
Name	NVARCHAR(100)	

Products

Column	Data Type	Key
Id	INT	PK
Name	NVARCHAR(100)	

Description	NVARCHAR(500)	
Price	DECIMAL(10,2)	
StockQuantity	INT	
CategoryId	INT	FK

Orders

Column	Data Type	Key
Id	INT	PK
CustomerId	INT	FK
OrderDate	DATETIME2	
TotalAmount	DECIMAL(10,2)	
Status	NVARCHAR(30)	

OrderItems

Column	Data Type	Key
Id	INT	PK
OrderId	INT	FK
ProductId	INT	FK
Quantity	INT	
UnitPrice	DECIMAL(10,2)	

3. Normalization

First Normal Form (1NF)

- Every column contains atomic values.
- No multiple values are stored in a single column.

Example:

✗ Wrong

Customer	Products
Ali	Laptop, Mouse

✓ Correct

Products are stored in a separate table (OrderItems).

Second Normal Form (2NF)

- Every non-key attribute depends on the entire primary key.
 - Product information is stored in the Products table instead of OrderItems.
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Third Normal Form (3NF)

- Every non-key attribute depends only on the primary key.
 - Customer information is stored only in the Customers table.
 - Orders store only CustomerId.
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4. Primary Keys

Table	Primary Key
Customers	Id
Categories	Id
Products	Id
Orders	Id
OrderItems	Id

5. Foreign Keys

Table	Foreign Key	References
Products	CategoryId	Categories(Id)
Orders	CustomerId	Customers(Id)
OrderItems	OrderId	Orders(Id)
OrderItems	ProductId	Products(Id)

6. Relationships

- One Customer can have many Orders.
 - One Category can have many Products.
 - One Order can have many OrderItems.
 - One Product can appear in many OrderItems.
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7. Data Types

Data	Type
Id	INT
Name	NVARCHAR(100)
Email	NVARCHAR(255)
Phone	NVARCHAR(20)
Description	NVARCHAR(500)
Price	DECIMAL(10,2)
StockQuantity	INT
Quantity	INT
OrderDate	DATETIME2
Status	NVARCHAR(30)

CUSTOMERS		
int	Id	PK
nvarchar	Name	
nvarchar	Email	
nvarchar	Phone	
nvarchar	Address	
datetime2	CreatedAt	

CATEGORIES		
int	Id	PK
nvarchar	Name	

ORDERS		
int	Id	PK
int	CustomerId	FK
datetime2	OrderDate	
decimal	TotalAmount	
nvarchar	Status	

PRODUCTS		
int	Id	PK
nvarchar	Name	
nvarchar	Description	
decimal	Price	
int	StockQuantity	
int	CategoryId	FK

ORDER_ITEMS		
int	Id	PK
int	OrderId	FK
int	ProductId	FK
int	Quantity	
decimal	UnitPrice	

